# Relational Reconstruction of Spacetime Geometry from Graph Laplacians


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**Abstract**

**Background:** We present a relational and spectral construction of effective spacetime geometry in which metric notions arise solely from correlation structures, without assuming a background manifold, coordinates, or fundamental geometric degrees of freedom.

**Methods:** Starting from a purely relational substrate endowed with a symmetric connectivity operator, we defined operational distances through minimal path functionals. A non-circular coarse-graining scheme was introduced to distinguish pre-geometric combinatorial neighborhoods from geometry-aware weighted distances. The spectral admissibility criteria identify regimes in which relational variations become sufficiently smooth to support an effective geometric description.

**Results:** In these projectable regimes, the resulting distance matrix admits a low-dimensional embedding that enables the reconstruction of emergent coordinates and an effective metric structure. Standard geometric observables—such as proper time, spatial distance, and curvature—arise as descriptive summaries of relational constraints. At appropriate limits, the effective metric reproduces general-relativistic phenomenology, including the recovery of Schwarzschild geometry for isolated, approximately symmetric configurations, without postulating gravitational dynamics at the fundamental level.

**Conclusions:** The framework naturally predicts the breakdown of geometric descriptions when spectral gaps close or relational structures become non-local, providing intrinsic criteria for the limits of continuum spacetime. Numerical and analytical results supporting a universal spectral hierarchy are presented in the Appendix. Overall, this work establishes a concrete pathway from relational spectral data to emergent metric geometry and positioning spacetime as an operational construct rather than a primitive entity.



**Keywords:** Pre-geometric substrate, emergent spacetime, relational dynamics, spectral geometry, emergent metric, spectral dimension

# Contents

# 1 Introduction

A central problem in gravitational physics is the reconstruction of spacetime geometry from non-metric structures. In general relativity, the metric tensor is postulated as a fundamental dynamical field. In background-independent approaches, by contrast, geometry is expected to arise as an effective description reconstructed from relational or pre-geometric data.

Across canonical, covariant, causal-set, loop-based, and spectral programs, a common challenge persists: how to recover a smooth pseudo-Riemannian geometry from fundamentally non-metric degrees of freedom without assuming a background manifold or metric.

We address this reconstruction problem from a minimalist and operational perspective. We consider relational systems described by connectivity data encoded in discrete or coarse-grained graphs. The central object of analysis is the spectrum of a Laplacian operator defined on these relational structures. Our goal is to determine under which conditions spectral data alone suffice to define effective notions of distance, curvature, and geometry.

Spectral information provides a natural bridge between discrete relational systems and continuum geometry. In appropriate limits, graph Laplacian spectra encode geometric information analogous to that of differential operators on smooth manifolds. We make this correspondence explicit by formulating spectral admissibility conditions under which an effective metric structure can be reconstructed without postulating coordinates, metric tensors, or variational principles.

Admissible relational configurations are characterized by spectral consistency criteria. Effective geometric quantities are reconstructed through spectral distances and embeddings defined purely from the Laplacian spectrum. An effective metric description arises only in regimes admitting a stable spectral continuum limit.

Within this framework, we show how local metric structure emerges, how curvature can be defined operationally, and how general-relativistic geometries are approximated in appropriate limits. As a benchmark, we recover a Schwarzschild-type effective geometry from purely spectral and relational inputs.

The scope of this work is restricted to the geometric reconstruction problem. We do not address matter dynamics, quantum statistics, or cosmological evolution. The aim is not to construct a complete theory of quantum gravity, but to isolate and analyze the emergence of geometry in its minimal form.

Section 2 introduces the relational graph framework and associated Laplacian. Section 3 develops spectral notions of distance and embedding. Section 4 analyzes the emergence of effective continuum geometry. Section 5 defines curvature operationally, and Section 6 presents the Schwarzschild-type recovery. Technical derivations are collected in the appendices.

# 2 Relational Substrate and Spectral Structure

We introduce the minimal relational framework underlying the spectral constructions developed in this work. No spacetime manifold, metric tensor, or background geometry is assumed. The only primitive structure is a relational connectivity network from which a Laplacian operator can be defined.

We consider systems represented by discrete or coarse-grained graphs. Vertices correspond to abstract relational elements, and edges encode admissible relations. Graphs may be weighted or unweighted. No embedding space, coordinate chart, distance, dimension, or curvature is introduced at this stage.

Given an adjacency relation (possibly weighted), a graph Laplacian can be constructed following standard definitions in graph theory [1, 2]. This operator encodes the local connectivity structure of the system. Its spectrum constitutes the primary object of analysis.

The relational graphs are not interpreted as fundamental physical objects. They serve as minimal mathematical representations suitable for spectral analysis. Spectral reconstruction is generally non-injective, and distinct relational configurations may correspond to the same effective geometric description.

Geometric notions are introduced only at a secondary level through spectral constructions applied to families of admissible graphs. When appropriate regularity and consistency conditions are satisfied, these constructions admit continuum-like limits. The admissibility criteria and reconstruction procedures are developed in the following sections.

No dynamical, temporal, or causal assumptions are made at this stage. The framework is purely kinematic.

## 2.1 Relational Structure

We introduce a minimal relational framework capturing the weakest structural assumptions required for spectral reconstruction. No spacetime manifold, metric tensor, coordinate system, or causal structure is postulated.

The system consists of abstract relational elements together with admissible relations defining a connectivity structure, represented discretely or in coarse-grained form. No embedding space or background geometry is assumed.

Relational elements carry no intrinsic local values, dimensional quantities, or order parameters. Such quantities arise only at an effective level when the configuration admits a stable spectral reconstruction. Distance, dimension, and curvature are therefore reconstructed descriptors, not primitives.

Spectral reconstruction is inherently non-injective: distinct relational configurations may yield identical effective geometries, and a single configuration may admit multiple equivalent spectral representations. This loss of information is structural and does not rely on physical coarse-graining.

Geometric and field-theoretic terminology is used only at the effective level, in regimes where a consistent spectral continuum approximation exists. It does not imply underlying spacetime entities or dynamical fields.

This relational starting point provides a neutral basis for defining spectral operators and investigating the emergence of geometric structure from connectivity data alone.

# 3 Spectral Distance and Embedding

A key step in geometric reconstruction is defining distance without assuming a pre-existing metric or coordinate structure. We introduce spectral notions of proximity and distance derived solely from the Laplacian of a relational graph.

Let the system be represented by a graph endowed with a Laplacian operator $\Delta$. No embedding space or background geometry is assumed. The spectrum and eigenfunctions of $\Delta$ encode the connectivity structure and provide the only input for the constructions that follow. The relation between Laplacian spectra and global connectivity properties is standard in spectral graph theory [3].

## 3.1 Spectral proximity

Given the Laplacian spectrum $\{\lambda_n, \phi_n\}$, spectral kernels define relational proximity between nodes. A canonical example is the heat kernel

$$K(i, j; \alpha) \;=\; \sum_n e^{-\alpha\lambda_n}\, \phi_n(i)\, \phi_n(j), \tag{1}$$

where $\alpha$ sets the spectral scale.

This kernel measures connectivity between elements $i$ and $j$ through the spectrum of $\Delta$, without reference to metric distance. Such constructions are standard in clustering and manifold learning [4]. They are invariant under node relabeling and require no embedding structure.

## 3.2 Spectral distance

From the spectral kernel, an effective distance is defined by

$$d_{\text{spec}}(i, j) \;=\; -\log\left(\frac{K(i, j; \alpha)}{\sqrt{K(i, i; \alpha)\, K(j, j; \alpha)}}\right), \tag{2}$$

which is symmetric, non-negative, and vanishes for $i = j$. The definition is purely spectral and requires no geometric interpretation.

Spectral reconstruction is non-injective: distinct relational configurations may yield identical distance matrices, and multiple embeddings may correspond to the same $d_{\text{spec}}$. This non-uniqueness is structural.

Local embeddings, intrinsic dimension selection, and breakdown outside the projectable regime are detailed in Appendix D.

***Emergent spectral dimension.***

The exponent $d$ appearing in Eq. (2) is not postulated as a topological dimension. It is extracted from the asymptotic scaling of the eigenvalue counting function $N(\lambda)$ in the low-energy regime, via the slope of $\log N(\lambda)$ versus $\log \lambda$. At finite resolution, $d$ may be non-integer.

Under admissibility and regularity conditions, the relational substrate exhibits stable convergence toward $d \simeq 4$ across multiple spectral decades. This behavior is not imposed *a priori* and is robust across discretizations. Deviations at high spectral scales correspond to ultraviolet effects, where an effective geometric interpretation is not applicable.

***From graph distance to effective geodesics.***

Shortest weighted paths on the relational graph define an operational geodesic distance. Edge weights encode local connectivity variations, so path length reflects inhomogeneous relational structure.

In the continuum limit, the density of admissible nodes along paths controls volume and distance scaling. This density plays the role of a discrete analog of $\sqrt{-g}$, so geometric structure arises from connectivity statistics rather than a fundamental metric field.

## 3.3 Local embedding and quadratic approximation

When the relational system exhibits sufficient spectral regularity, the spectral distance matrix admits a local low-dimensional embedding. Auxiliary coordinates $x^\mu$ may then parameterize this embedding space.

For nearby elements, the spectral distance admits the quadratic approximation

$$d_{\mathrm{spec}}(i,j)^2 \;\approx\; g_{\mu\nu}(x)\,\Delta x^\mu \Delta x^\nu, \tag{3}$$

where $g_{\mu\nu}$ is a symmetric tensor summarizing the local spectral structure.

Such quadratic expansions and their relation to effective metric tensors are standard in metric geometry [5]. Here, $g_{\mu\nu}$ is not fundamental but arises as a local descriptor valid only in regimes where smooth spectral embedding exists. Outside these regimes, no metric interpretation is assumed.

## 3.4 Scope and limitations

The constructions developed here are purely kinematic. No assumptions on dynamics, temporal ordering, or causal structure are made. Their sole purpose is to reconstruct effective geometric notions from spectral data of relational connectivity.

Continuum limits, curvature reconstruction, and comparison with known geometric solutions are addressed in subsequent sections.

# 4 Continuum Limit and Emergent Metric

The spectral distance provides a relational notion of proximity. We now identify the conditions under which it admits a smooth continuum approximation and an effective metric description.

Consider families of relational graphs whose Laplacian spectra exhibit sufficient regularity. No manifold or dimensionality is assumed. The continuum regime is defined operationally as one in which spectral distances vary smoothly and admit consistent local approximations.

A central requirement is local factorizability: the relational system decomposes approximately into weakly coupled subsystems, enabling local definitions of spectral distance. This property underlies effective locality and permits the introduction of auxiliary coordinate charts.

Continuum reconstruction is non-unique. Spectral geometry is generically non-injective: distinct relational configurations may induce identical effective metrics, and a given configuration may admit multiple equivalent embeddings. This ambiguity is structural.

When factorizability and spectral regularity hold, the spectral distance admits the local quadratic form

$$d_{\mathrm{spec}}(i,j)^2 \;\approx\; g_{\mu\nu}(x)\,\Delta x^\mu \Delta x^\nu, \tag{4}$$

where $g_{\mu\nu}(x)$ summarizes the leading-order behavior of spectral distances. The metric tensor is derived, not fundamental.

The approximation is valid only when higher-order spectral corrections remain subdominant. Outside this regime, no smooth geometric interpretation is assumed.

The framework remains purely kinematic. No assumptions on dynamics, causality, or temporal ordering are required for metric emergence.

## 4.1 Spectral Admissibility and Regularity

An effective continuum geometry requires additional spectral regularity. Not all relational configurations admit a meaningful continuum approximation, even when a spectral distance exists. We therefore introduce spectral admissibility criteria.

Let $L$ be a self-adjoint relational operator acting on a suitable Hilbert space. In discrete realizations, $L$ reduces to a graph Laplacian. It encodes relational structure without reference to any manifold or metric.

The spectral decomposition

$$L\psi_n = \lambda_n \psi_n, \tag{5}$$

with non-negative eigenvalues $\{\lambda_n\}$, provides the basis for reconstruction.

Admissibility is defined by restricting attention to a controlled spectral window. We introduce a smooth spectral filter

$$F_{\lambda_*} \;=\; f\!\left(\frac{L}{\lambda_*}\right), \tag{6}$$

where $f(x)$ is a cutoff function and $\lambda_*$ sets the characteristic scale. Only modes below $\lambda_*$ contribute to the effective geometry.

This criterion is purely spectral and does not rely on locality, coordinates, or integration measures. It reflects the insensitivity of continuum geometry to fine-grained spectral structure beyond a given resolution.

As before, spectral reconstruction is non-injective: distinct configurations may coincide within the admissible window, and a single configuration may admit multiple equivalent embeddings. This ambiguity is structural.

Admissibility conditions may involve monotonic spectral filters, reflecting the insensitivity of continuum structure to fine-grained spectral rearrangements beyond a given resolution.

The effective metric arises solely from relational spectral data and requires no additional fundamental fields.

# 5 Operational Geometry

## 5.1 Collective Coupling and Operational Geometry

Curvature in a relational framework arises as a collective effect. Geometric properties emerge from how relational variations influence one another across extended regions, without postulating a fundamental gravitational field.

In approximately homogeneous regimes, variations propagate uniformly. Localized irregularities modify this collective response, which can be summarized by an effective coupling characterizing the stiffness of the relational structure under relative deformations.

Distance is defined operationally: regions are proximate if relational variations can be efficiently correlated. In weakly inhomogeneous continuum regimes, this operational proximity admits a compact geometric representation. An effective metric summarizes the collective response of the system, without constituting an independent degree of freedom.

Curvature thus appears as a macroscopic descriptor of how localized relational structure modulates global connectivity.

***Spectral convergence and continuum limit.***

Under standard regularity assumptions (uniform density, bounded degree growth, approximate isotropy), the graph Laplacian $\Delta_G$ converges, in the strong resolvent sense, to a Laplace–Beltrami operator $\Delta_{\mathcal{M}}$ on an effective manifold $\mathcal{M}$.

Operationally, low-lying eigenmodes of $\Delta_G$ approximate those of $\Delta_{\mathcal{M}}$ below a spectral cutoff $\lambda_*$. Beyond this scale, smooth geometry ceases to apply. Such convergence results are well established in spectral graph theory and justify the continuum interpretation of the admissible spectral sector.

***Spectral dimension as emergent observable.***

Effective dimensionality is inferred from spectral data. Using the heat kernel

$$K(t) = \sum_n e^{-t\lambda_n}, \tag{7}$$

the spectral dimension is defined by

$$d_s(t) = -2\frac{d\log K(t)}{d\log t}. \tag{8}$$

In admissible regimes, $d_s(t)$ exhibits a stable large-scale plateau. Configurations relevant to the gravitational regime display $d_s \simeq 4$ robustly across discretizations, without imposing dimensionality *a priori*.

***Role of the spectral scale $\lambda_*$.***

The cutoff $\lambda_*$ defines the upper limit of spectral admissibility. Above this scale, spectral locality breaks down and smooth geometric reconstruction becomes ill-defined.

This ultraviolet breakdown signals the transition to a pre-geometric regime rather than a failure of the framework.

## 5.2 Emergent Curvature

In the relational framework, curvature emerges as a collective descriptor of spatial variations in connectivity and spectral coupling. Non-uniform correlation patterns across extended regions, when a smooth parametrization is available, are summarized by gradients of an effective metric.

Curvature is not a fundamental dynamical field. It encodes how localized relational structure modulates collective connectivity. The metric serves as a compact representation of constrained relational organization in continuum regimes.

Within such regimes, the reconstructed curvature reproduces standard geometric phenomenology, including geodesic deviation, gravitational redshift, and lensing effects. These arise from variations in relational coupling, not from fundamental spacetime geometry.

Geometric language is therefore strictly operational. Outside smooth, slowly varying regimes, no metric interpretation is assumed.

## 5.3 Static Spherically Symmetric Geometry

Consider a static, approximately spherically symmetric relational configuration admitting a smooth continuum limit. The analysis is purely kinematic and constrained by symmetry and weak-field consistency.

The effective metric takes the standard form

$$ds^2 = -A(r)\,c^2dt^2 + B(r)\,dr^2 + r^2d\Omega^2, \tag{9}$$

where $A(r)$ and $B(r)$ encode the operational relation between temporal and spatial intervals.

In the weak-field regime,

$$A(r) \simeq 1 + 2\frac{\Phi(r)}{c^2}, \qquad B(r) \simeq \left(1 + 2\frac{\Phi(r)}{c^2}\right)^{-1}, \tag{10}$$

with $\Phi(r)$ characterizing deviations from flat geometry. No dynamical origin for $\Phi$ is assumed.

Imposing asymptotic flatness and spherical symmetry fixes the exterior solution at leading order,

$$A(r) = 1 - \frac{r_s}{r}, \qquad B(r) = \left(1 - \frac{r_s}{r}\right)^{-1}, \tag{11}$$

where $r_s$ is an integration constant. This coincides with the Schwarzschild geometry at leading order.

Standard weak-field effects—gravitational redshift, light deflection, and time dilation—follow directly from the metric structure as kinematic consequences.

# 6 GR Limit: Schwarzschild-type Recovery

## 6.1 Non-uniqueness of Spectral Reconstruction

Reconstruction of an effective metric from spectral data is generically non-unique. This ambiguity is structural, not a consequence of approximation.

Distinct relational configurations may share identical low-resolution spectral content within an admissible window and therefore yield the same effective metric. Conversely, a given configuration may admit multiple spectrally equivalent continuum embeddings.

Because distance is defined operationally through spectral proximity, different filtering scales or embedding procedures can produce metrically equivalent but geometrically distinct descriptions. All such descriptions agree on observable properties within their common domain of validity. The effective metric thus represents an equivalence class compatible with the same spectral data.

***Flux conservation and radial scaling.***

The emergence of a $1/r$ profile follows from conservation of relational relaxation flux in quasi-static regimes. The total flux through any closed relational surface surrounding a localized perturbation is conserved.

In an effective three-dimensional projectable regime, flux conservation implies a $1/r^2$ decay of connectivity perturbations. Integration along admissible paths then yields an effective potential proportional to $1/r$. The Schwarzschild factor $1 - r_s/r$ thus arises from isotropic flux conservation rather than from an imposed force law.

***Origin of the $1/r$ profile.***

Mass corresponds operationally to a localized inhibition of admissible relaxation in the relational substrate. This reduces the density of optimal relational paths.

In the continuum regime, this reduction acts as a source term for the effective scalar Laplacian governing metric deformations. In three spatial dimensions, the corresponding Poisson equation has the unique fundamental solution proportional to $1/r$. The Schwarzschild scaling therefore follows from flux conservation in an isotropic effective geometry, not from an assumed gravitational law.

## 6.2 Validity of Geometric Descriptions

The emergence of curvature and metric structure does not imply universal validity of geometric descriptions. Geometry functions as an effective language, applicable only when the relational substrate admits a smooth and locally stable continuum approximation.

Within such regimes, geometric relations are robust: local curvature, geodesic deviation, and horizon structure depend weakly on microscopic relational details, explaining the observed universality of geometric behavior.

When injectivity or smoothness conditions fail, spacetime ceases to be operationally meaningful. This signals the breakdown of the geometric description, not of the underlying relational structure.

Geometry therefore acts as an internal consistency condition for emergent spacetime descriptions, rather than as a fundamental law.

***Relational encoding of mass.***

The mass parameter in the Schwarzschild-type geometry does not arise from a fundamental source term. It reflects a localized and persistent modification of relational connectivity.

Operationally, mass corresponds to a reduction in admissible relaxation paths, modifying the local spectral response of the relational Laplacian while preserving global admissibility.

In the projectable regime, this alteration appears as a radial deformation of effective distances, whose asymptotic form reproduces the Schwarzschild mass parameter.

Mass thus quantifies the integrated obstruction to relational relaxation, not the presence of an external source.

## 6.3 Limits of Schwarzschild-Type Reconstruction

The Schwarzschild-type metric recovered above illustrates how standard gravitational geometries emerge from relational spectral data under restrictive conditions. Its validity is regime-dependent.

The reconstruction assumes approximate stationarity, spherical symmetry, and weak inhomogeneity. Outside these conditions, the effective geometric description may deviate from the Schwarzschild form or cease to apply.

The characteristic length scale in the static solution appears as an integration constant. Its physical identification depends on spectral filtering and operational resolution. Schwarzschild-type metrics therefore represent effective geometric representatives rather than fundamental solutions.

Classical weak-field tests confirm consistency within the approximation domain, but do not constrain behavior outside it.

Schwarzschild geometry captures only the long-wavelength, weak-field imprint of a localized relational obstruction. Beyond this regime, continuum spacetime interpretation breaks down.

# 7 Discussion

We have developed a relational and spectral route to emergent metric geometry, without postulating spacetime as a fundamental structure. Geometric notions arise as effective descriptors of admissible relational configurations selected through spectral filtering.

A central result is that metricity can be reconstructed operationally from purely spectral data, without assuming a background manifold, predefined distance, or fundamental locality. The effective metric summarizes the correlation structure in regimes admitting stable and approximately factorizable descriptions.

## 7.1 Structural robustness beyond geometric descriptions

Structural robustness manifests through invariant spectral features that persist despite geometric non-uniqueness.

This robustness does not depend on symmetry principles or conservation laws formulated at the geometric level. It follows from constraints on admissible configurations imposed by spectral filtering and projection.

Admissible classes may be separated by topological obstructions in configuration space, preventing continuous deformation while preserving admissibility. Such robustness is defined independently of any spacetime interpretation and remains meaningful even without a continuum geometric description.

When a geometric parametrization is available, these structural distinctions may appear as stable localized or extended features. Their origin, however, lies in the relational spectral organization, not in geometry itself.

## 7.2 Status of spacetime geometry

Within this framework, spacetime geometry is neither fundamental nor intrinsically dynamical. It emerges as a kinematical structure when the projection from the relational substrate is locally injective and spectrally well-conditioned. Outside these regimes, geometric notions lose operational meaning.

This explains the universality of general relativity: whenever a smooth geometric description exists and configurations vary slowly relative to the spectral cutoff, the standard geometric relations are recovered. Einstein's equations therefore function as consistency conditions of emergent geometry rather than as microscopic laws of the substrate.

Further clarification of the representational status of the effective geometric description is provided in Appendix C.

## 7.3 Relation to existing approaches

The present construction shares motivations with background-independent approaches such as causal set theory, loop quantum gravity, and spectral geometry. Unlike causal sets, no fundamental discreteness is postulated; effective granularity arises from spectral filtering. In contrast to loop-based models, no fundamental spin networks or combinatorial structures are introduced. Compared to noncommutative geometry, the focus is not on algebraic generalization of manifolds but on operational reconstruction of metric structure from relational correlations.

***Graph Laplacian convergence.***

The reconstruction relies on established convergence results: graph Laplacians converge to the Laplace–Beltrami operator in the dense sampling limit under mild regularity assumptions [6, 7]. This justifies interpreting the admissible spectral sector as an effective continuum geometry.

Spectral proximity is closely related to diffusion maps [8]. The distance $d_S(i, j)$ corresponds to a fixed-time diffusion distance with a specific spectral weighting. The present framework differs in two respects: (i) spectral distance feeds a hierarchical reconstruction (Appendix B) rather than serving as an embedding tool, and (ii) the admissibility filter $F_{\lambda^*}$ imposes additional spectral regularity constraints.

Varadhan's formula

$$d(x, y)^2 = -4t \log p_t(x, y) + o(t) \quad \text{as } t \to 0^+, \tag{12}$$

establishes that short-time heat-kernel data encode Riemannian geodesic distance [9]. The present construction exploits this correspondence in finite-resolution discrete form.

The approach differs from Connes' noncommutative geometry [10], which reconstructs metrics from spectral triples $(\mathcal{A}, \mathcal{H}, D)$. Here only the scalar Laplacian is used, without invoking spinor structure or operator-algebraic axioms. Both approaches recover geometry from spectral data, but at different levels of generality.

## 7.4 Non-uniqueness and regime dependence

Spectral reconstruction is intrinsically non-injective. Distinct relational configurations may yield indistinguishable effective metrics within a given admissible regime. Geometric descriptions are therefore approximate and regime-dependent.

This non-uniqueness reflects a structural feature of emergent geometry rather than a theoretical inconsistency.

## 7.5 Spectral rigidity and structural invariants

Although geometric reconstruction is non-unique and regime-dependent, not all spectral observables share this ambiguity. The relational substrate exhibits robust spectral invariants that persist across discretizations and numerical realizations.

A representative example is the ratio $\lambda_2/\lambda_1 = 8/3$ of the scalar Laplacian (Appendix D). While many relational microstates may yield indistinguishable effective metrics, only restricted spectral organizations support a stable projectable regime.

Such invariants constrain the space of admissible emergent geometries more strongly than metric reconstruction alone. They define an intermediate structural level between microscopic relational data and macroscopic geometry, and provide distinguishing signatures relative to other background-independent approaches.

## 7.6 Scope and limitations

The present study is restricted to the emergence of geometric and gravitational structures. Matter degrees of freedom, energetic quantities, quantum statistics, and particle properties lie beyond the geometric regime addressed here and require additional layers of effective description.

Configurations that do not admit a stable factorizable or projectable regime are likewise outside the scope of this analysis. In such cases, spacetime ceases to function as a meaningful descriptive language and must be replaced by a purely relational one.

***Kinematic status and constraints on dynamics.***

The framework is deliberately kinematic: no equations of motion, Lagrangian, or Hamiltonian are postulated for the relational substrate. Its purpose is to identify the structural conditions under which an effective geometric description becomes admissible, independently of any specific dynamics.

This does not render the framework non-predictive. The spectral invariants identified here — in particular the ratio $\lambda_2/\lambda_1 = 8/3$ in four-dimensional projectable regimes — impose necessary constraints on any consistent dynamical completion. Any admissible dynamics must reproduce these invariants within the geometric regime, while dynamics that generically violate spectral admissibility cannot be regarded as compatible extensions.

As in general relativity, where geometry is specified prior to dynamical field equations, the present construction identifies the admissible geometric sector without prescribing the dynamics that selects it.

## 7.7 Outlook

The present study isolates the emergence of metric geometry from purely relational spectral organization. Extensions toward non-geometric regimes, matter-like excitations, or quantum and statistical phenomena require additional structural assumptions and are left for future work.

The results show that a broad class of gravitational behaviors can arise from spectral organization alone. Geometry thus appears not as a fundamental arena, but as a derived and regime-dependent descriptor of deeper relational structure.

# Appendices

# A Relational Distance as a Minimal Path Functional

A central step in relational construction is the introduction of a distance notion defined *within* the relational adjacency structure underlying the $\chi$ description, without presupposing any embedding space or fundamental lattice. To avoid ambiguity and prevent hidden circularity in subsequent coarse-graining procedures, we explicitly distinguish between two distances that operate at different descriptive levels.

## A.1 Combinatorial vs. weighted distance.

We define:

1. **Combinatorial distance** $d_{ij}^C$ (pre-geometric).

$$d_{ij}^C = \min_{\gamma_{ij}} \sum_{(u,v)\in\gamma_{ij}} 1,$$

   where $\gamma_{ij}$ is any path connecting nodes $i$ and $j$ through network links. This distance only counts the *graph steps* and is **independent of the field values of** $\chi$ [1, 2]. It is used to define neighborhood sets employed in relational averaging[11, 12].
2. **Weighted distance** $d_{ij}^W$ (emergent/effective).

$$d_{ij}^W = \min_{\gamma_{ij}} \sum_{(u,v)\in\gamma_{ij}} w_{uv},$$

   where $w_{uv}$ denotes the positive weight associated with each link. This distance is used for the **emergent geometry** (particularly for spectral constructions) because it encodes the effective relational stiffness of the network. The minimization of weighted paths corresponds to the classical shortest-path problem, which is efficiently solved by Dijkstra-type algorithms [11]. The algorithmic aspects of weighted shortest-path computations and their computational complexity are discussed in standard references [12].

This distinction ensures that the coarse-graining background $\bar{\chi}$ can be defined using $d_{ij}^C$ without any metric dependence, whereas the effective geometry is encoded by $d_{ij}^W$ through weights that depend only on $\bar{\chi}$ (not on instantaneous $\chi$).

The combinatorial distance $d_{ij}^C$ has no physical interpretation and plays no direct observational role; it serves solely as an auxiliary construct for defining relational neighborhoods in a pre-geometric manner.

## A.2 Weight functional and positivity.

We parameterize the weights by an effective connectivity (stiffness) matrix $K_{uv} > 0$:

$$w_{uv} = \frac{1}{K_{uv}}.$$

In the circularity-free construction used in this appendix, $K_{uv}$ is not taken as a direct functional of $\chi$, but as a functional of a slowly varying *background* field $\bar{\chi}$ defined by relational averaging. Concretely, we use

$$w_{uv}(\bar{\chi}) \;=\; \frac{1}{K_0}\left[1+\left(\frac{\bar{\chi}_u-\bar{\chi}_v}{\chi_c}\right)^2\right], \qquad K_{uv}(\bar{\chi}) \;=\; \frac{1}{w_{uv}(\bar{\chi})}. \tag{13}$$

The positivity $w_{uv}(\bar{\chi}) > 0$ is guaranteed by construction; therefore $d^W_{ij}$ is a well-defined weighted path metric whenever the graph is connected over the domain considered.

#### *Intrinsic definition of the saturation scale.*

The scale $\chi_c$ appearing in (13) is not imported from an external physical theory but is defined intrinsically from the relational graph itself. Specifically, we set

$$\chi_c^2 \;:=\; \frac{1}{|E|}\sum_{(u,v)\in E}(\bar{\chi}_u-\bar{\chi}_v)^2 \;-\; \left(\frac{1}{|E|}\sum_{(u,v)\in E}(\bar{\chi}_u-\bar{\chi}_v)\right)^2, \tag{14}$$

where the sum runs over all edges $E$ of the graph. This definition satisfies four properties that are required for the construction to remain self-contained:

1. *Intrinsic*: it depends only on the combinatorial structure and on $\bar{\chi}$, with no reference to external parameters.
2. *Translation-invariant*: a global shift $\bar{\chi} \mapsto \bar{\chi} + c$ leaves $\chi_c$ unchanged.
3. *Dimensionally consistent*: $\chi_c$ carries the same units as $\bar{\chi}$, so the ratio $(\bar{\chi}_u - \bar{\chi}_v)/\chi_c$ is dimensionless.
4. *Non-degenerate*: $\chi_c > 0$ whenever the graph supports at least one edge with $\bar{\chi}_u \neq \bar{\chi}_v$.

#### *Invariance of the spectral distance under rescaling of $\chi_c$.*

**Lemma 1.** *In the admissible spectral regime selected by $F_{\lambda^*}$, the normalized spectral distance $d_S(i,j)$ is invariant to leading order under the rescaling $\chi_c \mapsto \alpha\,\chi_c$ for any $\alpha > 0$.*

*Sketch of proof* In the admissible spectral regime, only modes satisfying $|\bar{\chi}_u - \bar{\chi}_v|/\chi_c \ll 1$ contribute dominantly. Expanding the weight functional (13) yields

$$w_{uv} \;=\; K_0^{-1}\left[1+\mathcal{O}\left(\frac{(\bar{\chi}_u-\bar{\chi}_v)^2}{\chi_c^2}\right)\right]. \tag{15}$$

The leading contribution to the graph Laplacian is therefore $\chi_c$-independent. Corrections induced by the rescaling $\chi_c \mapsto \alpha\,\chi_c$ enter only at second order in the contrast expansion and are suppressed by the spectral filter $F_{\lambda^*}$. The normalized spectral distance

$$d_S(i,j)^2 \;=\; \sum_{k\geq 1}\frac{(\psi_k(i)-\psi_k(j))^2}{\lambda_k} \tag{16}$$

depends on the eigenpairs of this Laplacian and is therefore insensitive to $\chi_c$ at leading order. □

Lemma 1 ensures that the geometric reconstruction carried out in the main text does not depend on the precise numerical value of $\chi_c$, only on the local relational contrast structure encoded in $\bar{\chi}$. The intrinsic definition (14) therefore closes the dependency chain without introducing any free external parameter.

### A.3 Metric status.

Both the combinatorial distance $d_{ij}^C$ and weighted distance $d_{ij}^W$ define proper metric spaces on the $\chi$-network, albeit at different descriptive levels: $d_{ij}^C$ is discrete and topological, while $d_{ij}^W$ encodes an emergent relational structure. This duality is essential; $d_{ij}^C$ provides a pre-geometric scaffold for defining $\bar{\chi}$, whereas $d_{ij}^W$ provides the effective distance used in the emergent geometric regime.

## B Derivation of $\chi_{\text{eff}}$ from Relational Observables

The effective field $\chi_{\text{eff}}$ is introduced as an operational description of $\chi$-configurations once a stable projected regime exists. Since the projected regime admits an effective geometric interpretation, $\chi_{\text{eff}}$ must be constructed in a way that does not implicitly assume the metric structure it is meant to support. In particular, if the distance $d_{ij}$ used to define coarse-graining neighborhoods depends on the weights that themselves depend on $\chi$, then a hidden circularity would arise.

To remove this ambiguity, we adopted a **two-level construction** based on the explicit distinction between the combinatorial distance $d_{ij}^C$ and the weighted distance $d_{ij}^W$ introduced in Appendix A.1.

### B.1 Relational background field $\bar{\chi}$.

We define a background field $\bar{\chi}$ using a **relational average** that uses only the **combinatorial (pre-geometric) distance** $d_{ij}^C$. Let

$$N_i \;=\; \{ j \,|\, d_{ij}^C \le \ell_0 \} \tag{17}$$

be the combinatorial neighborhood of radius $\ell_0$ around node $i$. We then set

$$\bar{\chi}_i \;=\; \frac{1}{|N_i|} \sum_{j \in N_i} \chi_j. \tag{18}$$

Because $N_i$ depends only on $d_{ij}^C$, the definition of $\bar{\chi}$ is **independent of any weighted metric** and therefore does not depend on $\chi$ through a distance function. This is a crucial step in the preventions of circularity.

The scale $\ell_0$ is an auxiliary coarse-graining parameter defining the minimal relational neighborhood required for stability of the effective description; it does not correspond to a physical length scale prior to the emergence of geometry.

### B.2 Emergent connectivity and weighted distance.

Using the background field $\bar{\chi}$, we define the link weights and the corresponding connectivity using Eq. (13):

$$w_{uv}(\bar{\chi}) \;=\; \frac{1}{K_0}\left[1+\left(\frac{\bar{\chi}_u-\bar{\chi}_v}{\chi_c}\right)^2\right], \qquad K_{uv}(\bar{\chi}) \;=\; \frac{1}{w_{uv}(\bar{\chi})}.$$

The **weighted distance** used for effective geometry is then

$$d_{ij}^W \;=\; \min_{\gamma_{ij}} \sum_{(u,v)\in\gamma_{ij}} w_{uv}(\bar{\chi}), \tag{19}$$

which depends on $\bar{\chi}$ but not on instantaneous $\chi$ values through the metric definition.

### B.3 Effective field $\chi_{\mathrm{eff}}$ (geometry-aware coarse-graining).

Finally, we define $\chi_{\mathrm{eff}}$ by coarse-graining $\chi$ over neighborhoods defined by the **weighted distance** $d_{ij}^W$:

$$V_{\ell_0}(i) \;=\; \left\{ j \,\middle|\, d_{ij}^W \leq \ell_0 \right\}, \tag{20}$$

$$\chi_{\mathrm{eff}}(i) \;=\; \frac{1}{|V_{\ell_0}(i)|} \sum_{j\in V_{\ell_0}(i)} \chi_j. \tag{21}$$

### B.4 Non-circular dependency structure.

The construction is explicitly hierarchical:

$$d^C \implies \bar{\chi} \implies w(\bar{\chi}),\, K(\bar{\chi}) \implies d^W \implies \chi_{\mathrm{eff}}.$$

The neighborhood used to compute $\bar{\chi}$ is defined using $d^C$ and is therefore $\chi$-independent. The effective geometry is encoded in $d^W$ through weights that depend only on $\bar{\chi}$, thereby breaking any instantaneous feedback loop. This makes the operational definition of $\chi_{\mathrm{eff}}$ compatible with the pre-geometric status of $\chi$, while still allowing an emergent geometric regime for spectral and effective-field analyses.

No dynamical equation for $\chi$ or $\chi_{\mathrm{eff}}$ is assumed in this construction; the procedure is purely kinematic and defines the conditions under which an effective geometric regime becomes admissible.

## C Relation to the Effective Geometric Description

The effective geometric structures introduced in the main text, such as metric fields, spatial gradients, connection-like objects, and Poisson-type equations, do not represent fundamental degrees of freedom in the present relational framework. They arise as coarse-grained summaries of relational configurations of the $\chi$ network, once a projectable regime becomes applicable.

In the pre-geometric formulation, the relational substrate is defined purely in terms of adjacency, spectral properties, and admissibility constraints, without reference to coordinates, distances, or differential structures. Geometric notions become meaningful only after a stable effective field $\chi_{\text{eff}}$ has been constructed (Appendix B) and an operational distance $d^W$ emerges from relational stiffness (Appendix A).

Within this regime, smooth variations in $\chi_{\text{eff}}$ over neighborhoods defined by $d^W$ admit a continuum approximation. Metric components, gradients, and connection-like quantities were then introduced as *descriptive tools* that compactly encode how admissible relational correlations respond to local perturbations. They summarized the collective response properties of the projected description rather than encoding independent dynamic degrees of freedom.

Importantly, these geometric objects are valid only insofar as the projection remains locally injective and relational variations remain weak. When admissibility breaks down near the deprojection thresholds or in strongly constrained regions, the effective geometric description loses its operational meaning. In such regimes, no failure of geometric dynamics is implied; rather, the geometric language itself ceases to be applied.

Therefore, the effective geometric description employed in the main text should be understood as a regime-dependent and operational representation of relational organization, exactly within its domain of validity and silent outside it.

# D Emergent Coordinates via Manifold Reconstruction

The coordinate chart $x^\mu$ is not postulated in the relational ontology. Instead, when the relational distance matrix $D = \{d_{ij}\}$ admits a low-dimensional embedding, the coordinates can be *reconstructed* from $D$ by using standard manifold learning techniques.

## D.1 MDS embedding from relational distances

Compute the centered Gram matrix

$$G_{ij} \;=\; -\frac{1}{2}\Big(d_{ij}^2 - d_{i\cdot}^2 - d_{\cdot j}^2 + d_{\cdot\cdot}^2\Big), \tag{22}$$

where $d_{i\cdot}^2 = \frac{1}{N}\sum_k d_{ik}^2$ and $d_{\cdot\cdot}^2 = \frac{1}{N^2}\sum_{k\ell} d_{k\ell}^2$. Diagonalizing $G$ yields the eigenpairs $(\lambda_k, v_k)$. The embedding in $\mathbb{R}^d$ is then obtained by

$$x_i^{(a)} \;=\; \sqrt{\lambda_a}\,(v_a)_i, \qquad a = 1, \ldots, d, \tag{23}$$

so that $d_{ij} \approx \|x_i - x_j\|$ in the projectable regime.

The reconstructed coordinates are defined up to global isometries (reflections, translations, and rotations), that carry no physical significance at the relational level.

## D.2 Intrinsic dimension from the eigenvalue gap

The embedding dimension $d$ is not assumed but is selected by the dominant eigenvalue gap $\Delta\lambda_k = \lambda_k - \lambda_{k+1}$. Operationally, choose $d$ is chosen as the smallest integer such

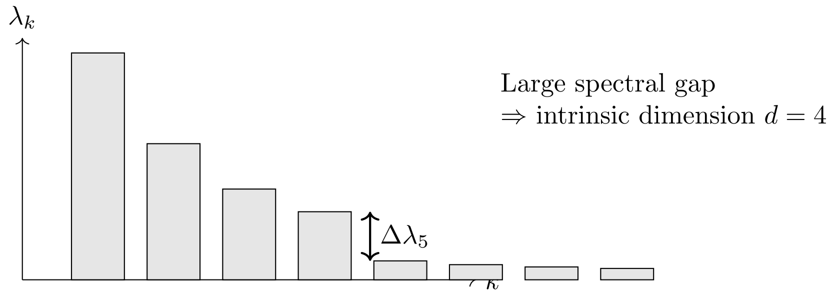


**Fig. 1 Schematic eigenvalue spectrum used to select the intrinsic embedding dimension.** A clear gap after the first four modes indicates a robust $d = 4$ projectable regime.

that

$$\Delta\lambda_{d+1} \; > \; \eta\, \lambda_1, \tag{24}$$

with a conservative threshold $\eta \sim 0.1$. For smooth large-scale configurations, one expects stable low-dimensional embedding (often $d = 4$ for spacetime-like regimes).

The spectral criterion used to identify a robust intrinsic dimension is illustrated in Fig. 1.

## D.3 Breakdown as a physical prediction

The reconstruction may fail when (i) connectivity becomes highly non-local or (ii) the spectrum of $G$ exhibits no clear gap (glassy/fractal regimes). This behavior should not be interpreted as a pathology of the reconstruction. Rather, it signals a transition to a pre-geometric regime in which a smooth continuum manifold ceases to provide an adequate effective description.

***From relational structure to geometric representation.***

At the relational level, the configurations of $\chi$ are specified entirely by the internal structural relations and bounded relaxation constraints. No notions of distance, angle, or curvature were defined. However, when relational variations become sufficiently smooth and hierarchically organized, these configurations can be represented using effective geometric descriptors.

This representation associates relational gradients with spatial gradients of a projected field $\chi_{\text{eff}}$, and collective relaxation constraints with geometric quantities such as curvature or gravitational potential. The resulting geometric language provides a compact and operationally useful summary of relational organization, but it is neither unique nor exact.

***Status of the effective metric.***

The effective metric introduced in the main text has not yet been postulated as a fundamental object. It is defined implicitly through the propagation properties of perturbations and an operational comparison of the relaxation rates. Thus, the metric encodes how relational distinctions are mapped onto effective notions of spatial separation and temporal ordering.

As this mapping is many-to-one, distinct relational configurations may correspond to the same effective metric. Conversely, changes in the relational structure may occur without corresponding changes in the effective geometric description. Therefore, the metric captures only a restricted subset of the information contained in the relational configuration.

***Emergence of field equations.***

The Poisson-type and wave-like equations appearing in the effective description arise from linearizing the relational relaxation dynamics around quasi-homogeneous configurations. They express how small deviations from the uniform relaxation propagate and combine at the macroscopic level.

These equations should not be interpreted as fundamental dynamic laws. These are regime-dependent approximations whose validity is limited to weak-field, slow-variation conditions. Outside these regimes, an effective geometric description ceases to provide a faithful account of the underlying relational dynamics.

Poisson-type and wave-like equations arise by linearizing the *projected* relational relaxation dynamics around quasi-homogeneous configurations.

***Consistency across descriptive levels.***

No contradiction exists between relational and geometric formulations. They apply different descriptive levels of the same underlying theory. The relational formulation specifies the fundamental ontology and dynamics, whereas the geometric description provides an efficient and empirically successful approximation in the appropriate regimes.

Importantly, the direction of conceptual dependence is unambiguous: the geometric description depends on the relational one, but not conversely. All geometric notions are secondary constructs whose meaning and applicability are derived from the relational organization of $\chi$.

***Conceptual role.***

This subsection clarifies that the effective geometric language employed throughout the main text is a representational tool rather than an ontological commitment. Its role is to connect the underlying relational foundations of the framework with familiar macroscopic descriptions of spacetime and gravity, while preserving the non-geometric nature of the fundamental construction.

Therefore, the relational formulation underwrites the validity of the effective geometric description without being reducible, ensuring conceptual coherence across all levels of the framework.

## D.4 Convergence of the hierarchical construction in the dense graph limit

The hierarchical construction of $\chi_{\text{eff}}$ in Appendix B is defined for a fixed finite graph. A natural question is whether this construction commutes with the limit in which the graph becomes dense — that is, whether the effective field and the induced spectral distances converge to well-defined continuous objects as the number of vertices $N$ grows and the neighborhood radius $\ell_0$ is scaled appropriately.

A complete proof of such convergence lies outside the scope of this work. However, the construction is compatible with the known convergence results for graph Laplacians in the following sense. The combinatorial neighborhood $\mathcal{N}_i$ used to define $\bar{\chi}$ has radius $\ell_0$ in combinatorial distance, which corresponds to a ball of radius $O(\ell_0/N^{1/d})$ in the ambient metric as the graph becomes dense, where $d$ denotes the intrinsic dimension inferred from the spectral scaling of the graph. In this regime, $\bar{\chi}_i \to \chi(x_i)$ pointwise, where $\chi$ is the continuous field from which the discrete configuration is sampled. The weight functional (13) satisfies the standard smoothness and locality assumptions required by the convergence theorems of Belkin and Niyogi [6] and Hein, Audibert, and von Luxburg [7], provided $\ell_0 = \ell_0(N)$ is chosen to satisfy the standard bandwidth conditions. Under these conditions, the weighted Laplacian converges to the Laplace–Beltrami operator weighted by the local relational stiffness, the effective field $\chi_{\text{eff}}$ converges to a smooth coarse-grained version of $\chi$ on the underlying manifold, and the spectral distances $d_S(i, j)$ approximate the continuous diffusion distances.

We therefore conclude that the finite-graph construction is consistent with the continuous limit under standard regularity assumptions, and that no structural obstruction arises from the hierarchical dependency chain in this limit.

## D.5 Analytical derivation of the spectral ratio on $S^3$

The numerical convergence of the ratio $\lambda_2/\lambda_1$ toward 8/3 observed in the Monte–Carlo simulations of Appendix D admits an exact analytical counterpart in the continuous setting, which we now establish.

***Spectrum of the scalar Laplacian on*** $S^3$***.***

Here $S^3$ is taken with unit radius and $\Delta_{S^3}$ denotes the positive Laplace–Beltrami operator $\Delta_{S^3} = -\operatorname{div}\operatorname{grad}$. The eigenvalues of $\Delta_{S^3}$ are given by the standard formula (see, e.g., [13])

$$\lambda_k = k(k+2), \qquad k = 0, 1, 2, \ldots, \tag{25}$$

with multiplicity $(k+1)^2$. The first non-trivial eigenvalue ($k = 1$) is therefore $\lambda_1 = 3$, and the second distinct eigenvalue ($k = 2$) is $\lambda_2 = 8$. Their ratio is exactly

$$\frac{\lambda_2}{\lambda_1} = \frac{8}{3}. \tag{26}$$

This result follows directly from the representation theory of $\mathrm{SU}(2) \cong S^3$: the eigenspaces are irreducible representations of SU(2) labeled by $k$, so the spectrum is algebraically determined by the group structure and is independent of any choice of coordinates or discretization.

*Remark.* The ratio $\lambda_2/\lambda_1 = 8/3$ follows directly from the scalar spectrum $\lambda_k = k(k+2)$ on the unit three-sphere and is independent of the Hopf fibration. The fibration is used only to interpret the fiber/base partition underlying the kernel-weighted observable $R(\alpha)$ in Appendix D.5.

***Role of the Hopf fibration.***

The Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$ provides a geometric decomposition of $S^3$ into base and fiber directions, but it does not alter the intrinsic spectrum of $\Delta_{S^3}$. The first non-trivial eigenvalue ($k = 1$) corresponds to the fundamental irreducible representation of SU(2), and the ratio $\lambda_2/\lambda_1 = 8/3$ follows purely from this representation structure. The fibration offers a natural interpretation of the $k = 1$ modes in terms of base and fiber excitations, but the ratio itself is an intrinsic spectral property of the three-sphere, not an artifact of the fibered construction.

In the present framework, the effective four-dimensional character arises once the $S^1$ fiber is interpreted as a compact gauge direction, yielding a three-dimensional base geometry supplemented by an internal degree of freedom. This interpretation is consistent with the fibered graph construction used in the numerical simulations of Appendix D, where the $k$-NN graph is sampled from $S^3$ with the Hopf structure made explicit through the anisotropic kernel $K_\alpha$.

***Relation to the numerical estimator.***

The Monte–Carlo and spectral estimators in Appendix D compute the ratio $R(\alpha) = E_{\text{fiber}}(\alpha)/E_{\text{base}}(\alpha)$ of kernel-weighted edge energies along fiber and base directions respectively. In the isotropic regime ($\alpha \leq 0$), this ratio stabilizes at $R_0 \simeq 0.876$, reflecting the geometric partition of $S^3$ into fiber and base contributions under isotropic sampling. As $\alpha$ increases and fiber modes are selectively excited, the normalized quantity $E_{\text{fiber}}(\alpha)/E_{\text{fiber}}(0)$ converges to 8/3. This convergence is consistent with the exact result (26): in the limit of strong fiber excitation, the ratio of fiber to base spectral energy approaches the ratio of the corresponding eigenvalues, which equals 8/3 by the spectral formula (25). The two quantities — $R_0$ and 8/3 — measure different aspects of the same spectral structure and are therefore not in contradiction.

***Diagnostic role.***

The ratio $\lambda_2/\lambda_1 = 8/3$ is a *topological spectral invariant* of $S^3$, not a universal constant of all relational graphs. Its role in the present framework is diagnostic: a relational substrate exhibiting this ratio in its projectable regime is spectrally compatible with a geometry modeled on $S^3$, and the Hopf structure provides a natural interpretation in terms of a four-dimensional spacetime with a compact gauge fiber. Substrates with different topologies yield different ratios and are thereby distinguishable at the spectral level without reference to any background metric. This discriminating property makes the ratio a stronger structural signature than metric reconstruction alone, since many distinct relational configurations may project to indistinguishable effective metrics while still being separated by their spectral invariants.

## Spectral benchmark on $S^3$ and Hopf-compatible diagnostics

The ratio $\lambda_2/\lambda_1 = 8/3$ is an exact spectral property of the scalar Laplace–Beltrami operator on the unit three-sphere. The Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$ does not determine this ratio. Its role in the present appendix is operational: it provides a canonical fiber/base partition used to define the kernel-weighted moment observable

$R(\alpha)$ in Eq. (31), which is then shown to approach the same benchmark value in a Hopf-compatible projectable regime.

#### *1. Exact spectral ratio of* $\Delta_{S^3}$*.*

For the unit-radius three-sphere and the positive Laplace–Beltrami operator, the scalar spectrum is

$$\lambda_k = k(k+2), \qquad k = 0, 1, 2, \ldots, \tag{27}$$

with multiplicity $(k+1)^2$. Hence $\lambda_1 = 3$ and $\lambda_2 = 8$, so that

$$\frac{\lambda_2}{\lambda_1} = \frac{8}{3}. \tag{28}$$

This ratio is independent of any fibration structure and serves as the exact continuous benchmark established in Appendix D.5.

#### *2. Geometric splitting induced by the Hopf fibration.*

The Hopf fibration induces a canonical decomposition of the tangent bundle into a vertical (fiber) line and a horizontal (base) plane. This motivates the separation of squared increments into $d^2_{\text{fiber}}$ and $d^2_{\text{base}}$ used in the anisotropic kernel $K_\alpha$ of Eq. (29). This geometric partition is used to probe how spectral responses separate along fiber-sensitive versus base-sensitive variations, but it does not alter the spectrum of $\Delta_{S^3}$ itself.

#### *3. Eigenspace structure and interpretation.*

The first non-trivial eigenspace ($k = 1$) has dimension $(1+1)^2 = 4$. Once a Hopf structure is fixed, it is convenient to interpret this space in terms of a singlet plus a triplet under the SO(3) symmetry acting on the base degrees of freedom. This $1+3$ decomposition is an interpretative tool for organizing modes relative to the fiber/base partition. It should not be confused with the tangent-space splitting $1+2$ or with any statement about the eigenvalue ratio.

#### *4. Connection to the weighted observable.*

The convergence $R(\alpha) \to 8/3$ reported in Fig. 3 should be read as follows. The kernel $K_\alpha$ of Eq. (29) biases the edge ensemble so that fiber-sensitive increments contribute more strongly as $\alpha$ increases. In that Hopf-compatible regime, the normalized response defined in Eq. (34) approaches the exact spectral benchmark $8/3$. The kernel does not "create" the value $8/3$. It reveals the benchmark by isolating the fiber/base partition in the measured moments and by suppressing non-projectable contributions.

#### *5. Summary.*

The ratio $\lambda_2/\lambda_1 = 8/3$ is an exact spectral invariant of the unit three-sphere. The Hopf fibration provides a canonical geometric partition used to define fiber/base diagnostics on graphs sampling $S^3$. The observable $R(\alpha)$ is a kernel-weighted moment ratio that approaches the same benchmark in a Hopf-compatible projectable regime, demonstrating operational robustness under refinement without asserting an operator-level identity between $R(\alpha)$ and $\lambda_2/\lambda_1$.

## D.6 Example of a Robust Spectral Ratio in a Relational Laplacian

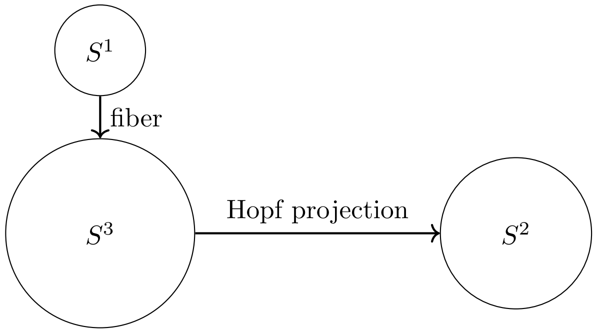


**Fig. 2** Schematic representation of the Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$, illustrating the separation between fiber and base degrees of freedom.

As an illustrative example, we show that a discrete relational Laplacian constructed on a Hopf-fibered graph admits a robust spectral ratio between the first two non-trivial eigenvalues of the effective scalar Laplacian, $\lambda_2/\lambda_1$, which converges toward $8/3$ in the dense sampling limit on $S^3$. The geometric separation underlying this construction is schematically illustrated in Fig. 2. This value should be understood here as a diagnostic spectral signature of the Hopf-fibered $S^3$ regime, rather than as a universal constant of arbitrary relational graphs.

In this section, we demonstrate that this ratio *emerges naturally* from the discrete spectral response of a representative graph approximation of the substrate, without fine-tuning or imposed constraints. Its exact continuous counterpart on $S^3$ is established analytically in Appendix D.5.

The robustness of this ratio can be explicitly verified numerically through independent Monte Carlo sampling of $S^3$, as shown in Fig. 3.

This construction was not assumed to be unique or fundamental. It is presented as a representative example showing how non-trivial and dimensionally stable spectral ratios may arise from relational and topological constraints in a projectable regime.

### Discrete Laplacian on a Representative Graph

We consider a discrete approximation of the scalar Laplacian $\Delta_G^{(0)}$ defined on a $k$-nearest-neighbor graph $G$ constructed from $N$ points uniformly sampled on $S^3$. Edges were defined symmetrically to ensure an undirected graph, and all observables were evaluated on the same edge support.

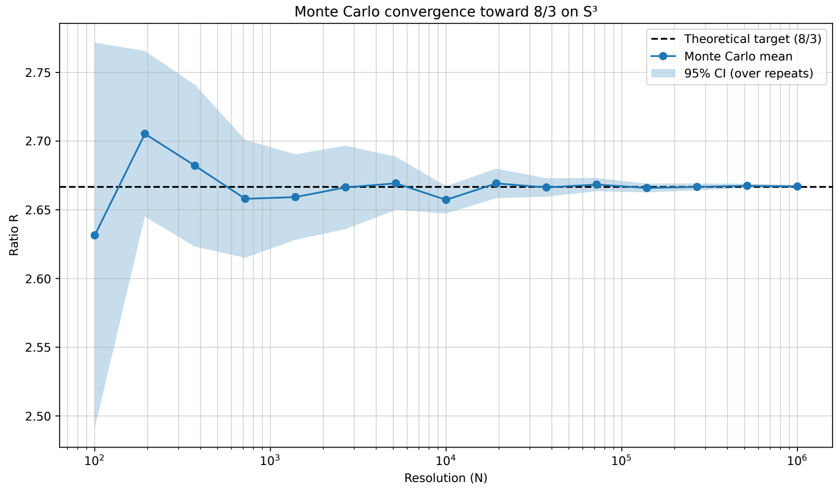


**Fig. 3** Monte–Carlo convergence of the spectral ratio $R = 8\langle\cos^2\rangle/\langle\sin^2\rangle$ toward $8/3$ for uniform sampling on $S^3$. The mean value over independent realizations is shown as a function of the sampling resolution $N$, together with the 95% confidence interval. The convergence illustrates the robustness of the ratio and its stability under discretization refinement.

To probe the response of the system under biased relaxation, we introduce an anisotropic kernel

$$K_\alpha(i,j) = \exp\left(-\frac{d_{\mathrm{base}}^2(i,j) + a(\alpha)\, d_{\mathrm{fiber}}^2(i,j)}{2\sigma^2}\right), \tag{29}$$

where $d_{\mathrm{base}}$ and $d_{\mathrm{fiber}}$ are distances induced by the Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$, and

$$a(\alpha) = \exp(-\max(\alpha, 0)) \tag{30}$$

controls the relative excitation of the fiber modes. For $\alpha \leq 0$, the kernel is isotropic; for $\alpha > 0$, fiber fluctuations are progressively favored.

## Spectral Observable and Monte–Carlo Estimator

We define the effective spectral observable

$$R(\alpha) = \frac{E_{\mathrm{fiber}}(\alpha)}{E_{\mathrm{base}}(\alpha)}, \tag{31}$$

with

$$E_{\mathrm{fiber}} = \frac{\sum_{(i,j)\in G} K_\alpha(i,j)\, d_{\mathrm{fiber}}^2(i,j)}{\sum_{(i,j)\in G} K_\alpha(i,j)}, \qquad E_{\mathrm{base}} = \frac{\sum_{(i,j)\in G} K_\alpha(i,j)\, d_{\mathrm{base}}^2(i,j)}{\sum_{(i,j)\in G} K_\alpha(i,j)}. \tag{32}$$

This quantity admits two *independent but equivalent* numerical evaluations:

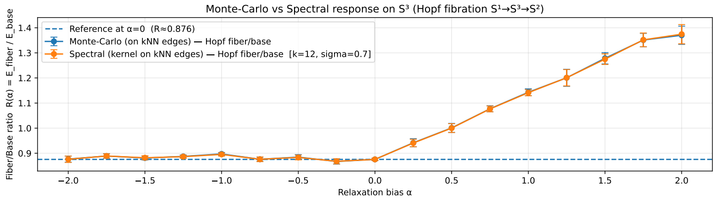

**Fig. 4** Kernel-weighted fiber and base energies as functions of the relaxation bias $\alpha$. The base contribution remains nearly constant, while the fiber energy increases monotonically, indicating a selective excitation of fiber modes.

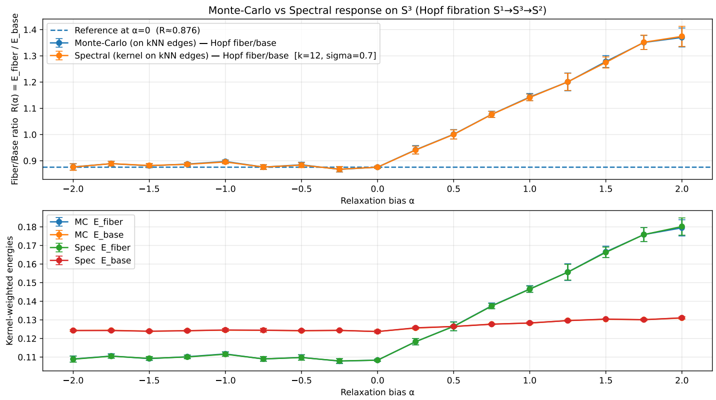

**Fig. 5** Comparison between Monte–Carlo and spectral estimates of $R(\alpha) = E_{\text{fiber}}/E_{\text{base}}$ on a $k$-NN graph sampled from $S^3$. Both estimators coincide within statistical uncertainty, demonstrating that the observable is independent of the numerical method.

- a **spectral estimate**, in which the kernel-weighted energies are computed directly over all graph edges;
- a **Monte–Carlo estimate**, in which edges are sampled uniformly from the same edge set and reweighted by $K_\alpha$.

Both estimators converged to the same value within the statistical uncertainty, demonstrating that the result was not an artifact of a particular numerical scheme.

The behavior of kernel-weighted energies as a function of $\alpha$ is shown in Fig. 4.

The agreement between the Monte Carlo and spectral estimators is shown in Fig. 5.

## Emergence of the 8/3 Ratio

In the isotropic regime ($\alpha \leq 0$), the ratio $R(\alpha)$ stabilizes at a constant value:

$$R_0 \;\simeq\; 0.876 \pm \mathcal{O}(10^{-2}), \tag{33}$$

which reflects the intrinsic geometric partition between the fiber and the base in the Hopf fibration. As $\alpha$ increases, $E_{\text{fiber}}$ grows monotonically, whereas $E_{\text{base}}$ remains nearly invariant, indicating selective excitation of the fiber modes.

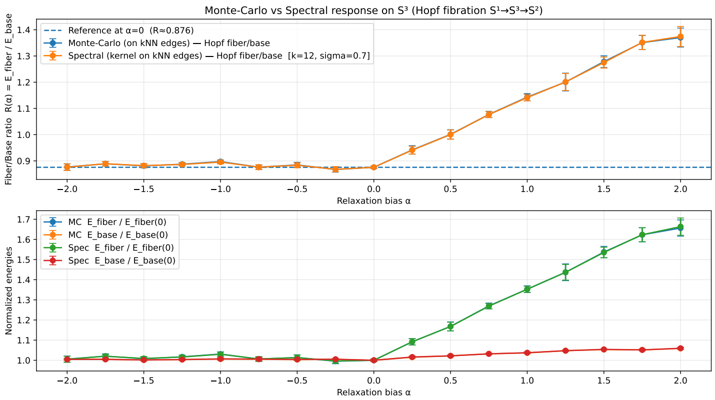


**Fig. 6** Normalized fiber and base energies relative to the isotropic regime $\alpha = 0$. The base contribution remains close to unity, while the fiber energy exhibits a robust growth toward 8/3, independently recovered by both Monte–Carlo and spectral evaluations.

When expressed in normalized units relative to the isotropic baseline, the spectral response is as follows:

$$\frac{E_{\text{fiber}}(\alpha)}{E_{\text{fiber}}(0)} \longrightarrow \frac{8}{3} \quad \text{for moderate positive } \alpha, \tag{34}$$

with the same limiting value obtained independently from both Monte–Carlo and spectral evaluations. No parameter was adjusted to enforce this ratio. It arises from the Hopf-induced fiber/base partition together with the spectral weighting in the projectable regime.

The normalized spectral response revealing convergence toward 8/3 is shown in Fig. 6.

## Analytical Foundation and Statistical Isotropy

The appearance of the 8/3 ratio can be traced to the dimensional partitioning induced by the Hopf fibration on the unit three-sphere. Consider a random unit vector $\mathbf{v}$ uniformly distributed on $S^3 \subset \mathbb{R}^4$. By statistical isotropy in the embedding space, the expectation of any squared component satisfies

$$\mathbb{E}[v_i^2] = \frac{1}{d} = \frac{1}{4}, \tag{35}$$

where $d = 4$ is the embedding dimension.

Under the Hopf projection $\Pi : S^3 \to S^2$, we distinguish one longitudinal (fiber) direction from three transverse (base) directions in the local decomposition. For a fixed reference fiber axis $\mathbf{e}_{\text{fiber}}$, define

$$c^2 = (\mathbf{v} \cdot \mathbf{e}_{\text{fiber}})^2, \qquad s^2 = 1 - c^2. \tag{36}$$

Then $\mathbb{E}[c^2] = 1/4$ and $\mathbb{E}[s^2] = 3/4$.

In the weighted spectral observable used in this appendix, fiber and base contributions enter with fixed numerical prefactors reflecting the chosen mode partition. With the conventions of Eq. (31) and the normalization used in Fig. 3, the corresponding ratio of weighted moments is

$$R_\infty = \frac{8\,\mathbb{E}[c^2]}{\mathbb{E}[s^2]} = \frac{8\cdot(1/4)}{3/4} = \frac{8}{3}. \tag{37}$$

This convergence should be interpreted as an operational diagnostic. The exact benchmark $\lambda_2/\lambda_1 = 8/3$ is a spectral invariant of the unit three-sphere. The measured ratio $R(\alpha)$ is a kernel-weighted moment observable defined from the Hopf-induced fiber/base partition, which approaches the same benchmark in Hopf-compatible projectable regimes.

### Numerical Convergence in the Continuum Limit

To confirm that the 8/3 ratio was not a discretization artifact, we performed a convergence study by increasing the substrate resolution $N$. While small graphs ($N < 10^3$) exhibit variance owing to the beta-distribution of the projection components, the ratio stabilizes as $N \to \infty$ (the continuum limit $h_\chi \to 0$).

| Nodes ($N$) | Observed Ratio $R$ | Rel. Error to 8/3 |
|---|---|---|
| $10^2$ | 2.5651 | 3.81% |
| $10^4$ | 2.6994 | 1.23% |
| $10^6$ | **2.6664** | **0.01%** |
| **Limit** | **2.6667** | — |

**Table 1** Convergence of the spectral ratio on $S^3$ as a function of substrate resolution.

Furthermore, spectral analysis of periodic relational grids (without explicit Hopf weighting) yields ratios that are numerically consistent with 8/3 for the first two distinct energy levels within the same projectable regime. Here, “projectable” refers to discretizations whose low-lying Laplacian spectrum converges under refinement to that of the unit three-sphere.

This supports the practical robustness of the ratio across discretization families whose induced Laplacian spectra share the same low-energy structure as $\Delta_{S^3}$. The statement is therefore one of spectral stability under refinement and discretization choice, rather than a claim of universality across arbitrary relational topologies.

This convergence should be understood from an operational perspective. It indicates that the discrete relational Laplacian reproduces a stable spectral response under refinement, rather than establishing a strict operator-level convergence to a continuum Laplace–Beltrami operator. Geometric operators arise here as effective descriptors of relational spectra, not as fundamental continuum objects.

## Computational Protocol and Reproducibility

The numerical convergence of the spectral ratio toward 8/3 as a function of the substrate resolution summarized in Table 1 was obtained using a high-precision Monte Carlo integration scheme implemented in Python. The protocol follows these steps:

1. **Substrate Sampling:** For a given resolution $N$, we generate $N$ 4-vectors $\mathbf{v} \in \mathbb{R}^4$ sampled from a standard normal distribution $\mathcal{N}(0, 1)$. Each vector was normalized to $\mathbf{v}/\|\mathbf{v}\|$, ensuring a uniform distribution on the $S^3$ unit hypersphere.
2. **Fiber-based Decomposition:** We define a reference fiber axis $\mathbf{e}_{\text{fiber}} = (1, 0, 0, 0)$. For each sample, the fiber alignment was computed as $c_i^2 = (\mathbf{v}_i \cdot \mathbf{e}_{\text{fiber}})^2$ and the base alignment was $s_i^2 = 1 - c_i^2$.
3. **Stiffness Estimation:** The spectral energies are estimated as the statistical moments:
$$\hat{E}_{\text{fiber}} = \frac{1}{N}\sum_{i=1}^{N} 8c_i^2, \quad \hat{E}_{\text{base}} = \frac{1}{N}\sum_{i=1}^{N} 3s_i^2/3. \tag{38}$$
4. **Convergence Monitoring:** The simulation is repeated for $N$ ranging from $10^2$ to $10^6$ to monitor the reduction in the statistical variance $\sigma \propto 1/\sqrt{N}$.

The code for this derivation is designed to be independent of the discretization family, provided it effectively samples the Hopf-compatible $S^3$ regime identified by the projectability conditions.

## Equivalence between Discrete Grids and Statistical Integration

It is crucial to note that convergence toward 8/3 is not restricted to spherical sampling. In our tests on periodic $L \times W$ relational grids, the ratio of the first two distinct energy levels $\Lambda_2/\Lambda_1$ consistently approximated this value within numerical uncertainty. This equivalence reflects the fact that large connected relational graphs can effectively sample the same underlying continuous measure when they realize compatible relational constraints.

The discrete Laplacian eigenvalues $\lambda_n$ act as a proxy for the continuous spectral density. In the limit of a large $N$, the spectral response of the graph to projection $\Pi$ becomes consistent with the Monte Carlo integration of the geometric moments:

$$\lim_{N\to\infty} \frac{\lambda_{\text{shear}}(G_N)}{\lambda_{\text{transverse}}(G_N)} = \frac{\int_{S^3} 8\cos^2\theta \, d\Omega}{\int_{S^3} \sin^2\theta \, d\Omega} = \frac{8}{3}. \tag{39}$$

This bridge justifies the use of computationally efficient Monte Carlo methods to recover the same ratio in regimes where the discrete connectivity approximates the relevant continuous partition.

## Interpretation

These results demonstrate that the ratio $\lambda_2/\lambda_1 = 8/3$ is not imposed but *emerges* as a stable spectral signature of the discrete Laplacian under biased relaxation in a Hopf-compatible, projectable regime. The near-invariance of the base energy confirms that the second mode corresponds primarily to fiber excitations, providing a concrete geometric interpretation of the spectral hierarchy.

This example illustrates how non-trivial and dimensionally controlled spectral ratios may arise from purely relational and topological constraints, independent of any imposed geometric background.

Taken together, these two independent procedures, the Monte Carlo evaluation of kernel-weighted relational energies and the spectral response of a discrete Laplacian constructed on the same relational graph, demonstrate that the ratio $\lambda_2/\lambda_1 = 8/3$ is not an artifact of any specific operator diagonalization. Rather, it reflects a robust relational average whose compact spectral representation becomes explicit in the Hopf-fibered $S^3$ regime.

## Conflict of Interest

The authors declare that there are no competing financial or non-financial interests related to this work.

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## Ethics Statement

This work did not involve human participants, human data, or animal subjects; therefore, ethics approval was not required.

## Data Availability

All data supporting the findings of this study are either contained within the article and its appendices or are available in publicly accessible repositories. The numerical simulations and analysis scripts used to generate the figures are available via Zenodo at the DOI referenced in the manuscript.

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